

Supplementary Material: Continuous Collision Detection Trust Regions for Clothing Simulation

ANONYMOUS AUTHOR(S)
SUBMISSION ID: 1107

ACM Reference Format:

Anonymous Author(s). 2026. Supplementary Material: Continuous Collision Detection Trust Regions for Clothing Simulation. 1, 1 (May 2026), 2 pages. <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

0.1 Incremental Potential and Energy Decomposition

We adopt the standard incremental potential formulation [4]. A backward Euler step reduces to the minimization

$$\Phi(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_{\mathbf{M}}^2 + (\Delta t)^2 PE(\mathbf{x}), \quad (1)$$

where $\|\mathbf{v}\|_{\mathbf{M}}^2 = \mathbf{v}^T \mathbf{M} \mathbf{v}$, $\hat{\mathbf{x}} = \mathbf{x}^n + \Delta t \mathbf{v}^n$ is the inertial extrapolation, and $\mathbf{M} = \text{diag}(m_1 \mathbf{I}_3, \dots, m_N \mathbf{I}_3)$ is the lumped mass matrix. We decompose the total potential as

$$PE(\mathbf{x}) = PE^e(\mathbf{x}) + PE^b(\mathbf{x}) + PE^g(\mathbf{x}) + PE^{\text{bar}}(\mathbf{x}) + PE^{\text{pin}}(\mathbf{x}), \quad (2)$$

corresponding respectively to corotated elastic stretching, discrete shell bending, gravity, IPC barrier contact, and pinning constraints. We now briefly state each term.

Corotated elasticity. On each triangle with rest-shape matrix $\mathbf{D}_m \in \mathbb{R}^{2 \times 2}$, deformed segment matrix $\mathbf{D}_s \in \mathbb{R}^{3 \times 2}$, and deformation gradient $\mathbf{F} = \mathbf{D}_s \mathbf{D}_m^{-1}$, we use the corotated energy density [5]

$$\psi(\mathbf{F}) = \mu \|\mathbf{F} - \mathbf{R}\|_F^2 + \frac{\lambda}{2} (J - 1)^2, \quad (3)$$

where $\mathbf{F} = \mathbf{R}\mathbf{S}$ is the polar decomposition, $J = \sqrt{\det(\mathbf{F}^T \mathbf{F})}$, and μ, λ are Lamé parameters. The element elastic energy is $E_t = A_t \psi(\mathbf{F})$ with rest area $A_t = \frac{1}{2} |\det \mathbf{D}_m|$, and the total stretching energy is $PE^e = \sum_t E_t$.

Bending. Following Grinspun et al. [3], the discrete shell bending energy for a hinge segment e with dihedral-angle complement θ and rest value $\bar{\theta}$ is

$$w_e = k_B c_e (\theta - \bar{\theta})^2, \quad c_e = \frac{|\bar{e}|}{h_e}, \quad (4)$$

where k_B is the bending stiffness and c_e is a geometric coefficient depending on the rest segment length and opposite heights. The total bending energy is $PE^b = \sum_e w_e$.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2026 Copyright held by the owner/author(s). Publication rights licensed to ACM. ACM XXXX-XXXX/2026/5-ART <https://doi.org/10.1145/nnnnnnnn.nnnnnnnn>

Gravity and pinning. Gravity contributes a linear potential $PE_i^g(\mathbf{x}_i) = -m_i \mathbf{g}_{\text{grav}}^T \mathbf{x}_i$ at each node. Pinned nodes are enforced through a quadratic penalty $PE_i^{\text{pin}}(\mathbf{x}_i) = \frac{k_{\text{pin}}}{2} \|\mathbf{x}_i - \mathbf{x}_i^{\text{pin}}\|^2$ with $k_{\text{pin}} \gg 1$.

IPC barrier. Non-penetration between deformable primitives is enforced by the IPC log-barrier [4]. For a primitive pair with gap $\delta > 0$, the barrier function is

$$b(\delta; \hat{d}) = \begin{cases} -(\delta - \hat{d})^2 \ln(\delta/\hat{d}), & 0 < \delta < \hat{d}, \\ 0, & \delta \geq \hat{d}, \end{cases} \quad (5)$$

and the total barrier energy is

$$PE^{\text{bar}}(\mathbf{x}) = \kappa_{\text{bar}} \sum_{(p,q)} b(\delta_{pq}; \hat{d}), \quad (6)$$

where κ_{bar} is the barrier stiffness and the sum ranges over all active vertex-triangle and edge-edge pairs.

0.2 Signed-Distance Penalty Energy

We additionally enforce non-penetration with respect to rigid obstacles through a penalty energy composed from the signed distance function and a piecewise-linear Heaviside. Let $\Omega \subset \mathbb{R}^3$ denote the obstacle region with signed distance function $\phi(\mathbf{x})$, negative inside and positive outside. Define the piecewise-linear Heaviside

$$H(z) = \begin{cases} 1, & z < 0, \\ (\epsilon - z)/\epsilon, & 0 \leq z \leq \epsilon, \\ 0, & z > \epsilon, \end{cases}$$

so that H transitions linearly from full penetration to zero over a band of width ϵ . The per-node penalty energy is

$$PE_i^{\text{sdf}}(\mathbf{x}_i) = \frac{k_{\text{sdf}}}{2} [H(\phi(\mathbf{x}_i))]^2, \quad (7)$$

where k_{sdf} is a stiffness parameter. Nodes with $\phi(\mathbf{x}_i) > \epsilon$ contribute zero energy and are skipped.

0.3 Local Newton Subproblem

The Gauss-Seidel strategy updates nodes sequentially through local implicit subproblems. Each local step incorporates the inertia, elastic, gravitational, pinning, and active barrier contributions associated with the current node.

For node i , the local first-order optimality condition is

$$\nabla_{\mathbf{x}_i} \Phi(\mathbf{x}) = m_i(\mathbf{x}_i - \hat{\mathbf{x}}_i) + (\Delta t)^2 (\nabla_{\mathbf{x}_i} PE^e + \nabla_{\mathbf{x}_i} PE^b + \nabla_{\mathbf{x}_i} PE^{\text{bar}} + \nabla_{\mathbf{x}_i} PE^{\text{pin}} - m_i \mathbf{g}_{\text{grav}}) = \mathbf{0}, \quad (8)$$

where the pinning term is present only if node i is pinned. Linearizing about the current $\mathbf{x}_i$ and solving for the correction $\delta_i \in \mathbb{R}^3$ gives

$$\mathbf{H}_{ii} \delta_i = \mathbf{g}_i, \quad (9)$$

where $\mathbf{g}_i = \nabla_{\mathbf{x}_i} \Phi(\mathbf{x})$ and

$$\mathbf{H}_{ii} = m_i \mathbf{I}_3 + (\Delta t)^2 (\mathbf{H}_{ii}^e + \mathbf{H}_{ii}^b + \mathbf{H}_{ii}^{\text{bar}} + \mathbf{H}_{ii}^{\text{pin}}) \in \mathbb{R}^{3 \times 3}. \quad (10)$$

Here $\mathbf{H}_{ii}^e$ is the diagonal block of the elastic Hessian summed over all triangles incident to node i , $\mathbf{H}_{ii}^b$ is the diagonal block of the bending Hessian summed over all hinge segments incident to node i , $\mathbf{H}_{ii}^{\text{bar}}$ is the diagonal block of the barrier Hessian summed over all active barrier pairs involving node i , and $\mathbf{H}_{ii}^{\text{pin}} = k_{\text{pin}} \mathbf{I}_3$ if i is pinned and zero otherwise. The node position is then updated as

$$\mathbf{x}_i \leftarrow \mathbf{x}_i - \omega_i \delta_i, \quad (11)$$

where $\omega_i \in (0, 1]$ is a step size determined by linear CCD in Section 0.5.

0.4 Initial Guess

The initial iterate $\mathbf{x}^{(0)}$ must be collision-free and connected to $\mathbf{x}^n$ by a linear trajectory so that CCD queries are well defined.

The inertial term $\frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_{\mathbf{M}}^2$ in the incremental potential (1) is minimized at the inertial extrapolation $\hat{\mathbf{x}} = \mathbf{x}^n + \Delta t \mathbf{v}^n$, which is typically the dominant contribution to Φ . A natural initial guess is therefore to advance toward $\hat{\mathbf{x}}$ as far as collision constraints allow. Since all nodes move simultaneously along this trajectory, the coplanarity polynomial for each candidate primitive pair is a full cubic in t [1]. For every candidate pair we compute the earliest collision time $t^* \in [0, 1]$ and set

$$\omega_0 = \min(1, \min_{\text{all pairs}} 0.9 t^*), \quad (12)$$

$$\mathbf{x}^{(0)} = \mathbf{x}^n + \omega_0 \Delta t \mathbf{v}^n. \quad (13)$$

This places the initial guess at the collision-free configuration that minimizes the inertial energy along the line from $\mathbf{x}^n$ to $\hat{\mathbf{x}}$, so the solver only needs to resolve the potential energies.

0.5 Linear CCD

Each local Newton update must be filtered to ensure collision freedom. We use CCD based on coplanarity tests [1] to determine the maximal admissible step size. Candidate primitive pairs are identified via swept axis-aligned bounding boxes organized in bounding volume hierarchies [2, Section 4.2].

Since only a single node moves per local step, the coplanarity polynomial simplifies significantly. Given four points interpolating as $\mathbf{x}_i(t) = \mathbf{x}_i^0 + t \mathbf{d}_i$ with $t \in [0, 1]$, define the edge vectors $\mathbf{p} = \mathbf{x}_1 - \mathbf{x}_0$, $\mathbf{q} = \mathbf{x}_2 - \mathbf{x}_0$, $\mathbf{r} = \mathbf{x}_3 - \mathbf{x}_0$. The scalar triple product $f(t) = (\mathbf{p}(t) \times \mathbf{q}(t)) \cdot \mathbf{r}(t)$ is cubic in general, but when only one vertex carries nonzero displacement, at most one of $\mathbf{p}, \mathbf{q}, \mathbf{r}$ depends on t , so

$$f(t) = ct + d, \quad t^* = -d/c. \quad (14)$$

For example, when the query vertex $\mathbf{x}_0$ moves with displacement $\mathbf{d}_0$, the coefficients are $d = (\mathbf{p}_0 \times \mathbf{q}_0) \cdot \mathbf{r}_0$ and $c = -(\mathbf{p}_0 \times \mathbf{q}_0) \cdot \mathbf{d}_0 + (\mathbf{p}_0 \times \mathbf{d}_0) \cdot \mathbf{r}_0 + (\mathbf{d}_0 \times \mathbf{q}_0) \cdot \mathbf{r}_0$, where $\mathbf{p}_0, \mathbf{q}_0, \mathbf{r}_0$ are the edge vectors at $t=0$. A candidate t^* is accepted only if $|c| > \epsilon$, $t^* \in (0, 1]$, and the contact point lies inside the primitive (positive barycentric coordinates for vertex-triangle; both edge parameters in $[0, 1]$ for edge-edge). The step is clamped to $\hat{t} = 0.9 \min t^*$ to maintain a safety margin.

REFERENCES

- [1] Robert Bridson, Ronald Fedkiw, and John Anderson. 2002. Robust Treatment of Collisions, Contact and Friction for Cloth Animation. *ACM Transactions on Graphics* 21, 3 (2002), 594–603. <https://doi.org/10.1145/566654.566623>
- [2] Christer Ericson. 2004. *Real-Time Collision Detection*. Morgan Kaufmann, San Francisco, CA.
- [3] Eitan Grinspun, Anil N. Hirani, Mathieu Desbrun, and Peter Schröder. 2003. Discrete Shells. In *Proceedings of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA '03)*. Eurographics Association, San Diego, CA, 62–67.
- [4] Minchen Li, Zachary Ferguson, Teseo Schneider, Timothy Langlois, Denis Zorin, Daniele Panozzo, Chenfanfu Jiang, and Danny M. Kaufman. 2020. Incremental Potential Contact: Intersection- and Inversion-Free, Large-Deformation Dynamics. *ACM Transactions on Graphics* 39, 4 (2020), 49:1–49:20. <https://doi.org/10.1145/3386569.3392425>
- [5] Alexey Stomakhin, Russell Howes, Craig Schroeder, and Joseph M. Teran. 2012. Energetically Consistent Invertible Elasticity. In *Proceedings of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA '12)*. Eurographics Association, Lausanne, Switzerland, 25–32.